Fiscal Year:	FY 2022	**Task Last Updated:**	FY 11/30/2022
PI Name:	Blanco, Gustavo Ph.D.		
Project Title:	Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	02/09/2018	**End Date:**	03/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 3/31/2021 per F. Hernandez/ARC (Ed., 4/7/2020) NOTE: Extended to 3/31/2020 per F. Hernandez/ARC and NSSC information (Ed. 5/15/19) NOTE: Extended to 8/31/2019 per F. Hernandez/ARC (Ed. 10/29/18)		
Key Personnel Changes/Previous PI:	January 2018: Former Principal Investigator was Joseph Tash, Ph.D., who retired January 2018. Dr. Tash remains as consultant on the project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0522		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE (June 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" (grant number NNX13AM62G) due to retirement of former Principal Investigator Joseph Tash, Ph.D. Originally the grant had continued with same grant number as noted in January 2018 below note; in February 2018 new grant number 80NSSC18K0522 with new period of performance was awarded. NOTE (January 2018): Continuation of "Space Flight-Altered Motility Activation and Fertility-Dependent Responses in Sperm from Sea Urchin and Rodents" with the same grant number NNX13AM62G due to retirement of former Principal Investigator (PI) Joseph Tash, Ph.D. NOTE also that in 2016 with predecessor PI Tash, work on the grant was amended to include cryopreserved bovine and human semen, while eliminating the use of sea urchin and rat species. Sperm motility and acquisition of the capacity to fertilize are the most fundamental components of male reproduction necessary for species survival on Earth or in space. Previous space shuttle flight experiments that we performed demonstrated that the signal transduction (i.e., m protein phosphorylation) required for activation of sea urchin sperm occur more rapidly in microgravity than at 1G. In contrast, we found that signaling in sperm triggered by egg chemotactic factors is delayed in microgravity compared to 1G. No flight-capable microscope was available during those experiments to determine whether the signaling changes were in fact coupled to differences in motility under the microgravity versus 1G conditions. Our hypothesis is 1) that sperm motility (sea urchin and mammalian) is activated more rapidly in microgravity than on Earth, and 2) that the response of sea urchin and mammalian sperm (motility hyperactivation, and morphological changes) to fertilization activating conditions, as well as the underlying metabolic and regulatory mechanisms, are delayed or blocked in microgravity. In this regard, if the corresponding activation and changes in sperm motility do not occur, then fertilization cannot occur and reproduction will fail. Thus, it is essential to assess whether the critical component of sperm function, motility, and the underlying signalling and metabolic support are altered in space flight conditions. We propose to use existing flight hardware to activate sea urchin and rodent sperm in microgravity, and quantify multiple parameters of motility and morphology, combined with detailed metabolomic pathway analysis. Previously utilized methods for preparing inactive sea urchin sperm for subsequent activation in space flight will be adapted for mouse and rat sperm. We will also test whether activating peptides from sea urchin eggs, or mammalian capacitating conditions trigger temporal or magnitude changes in sperm motility, morphology, as well as essential regulatory glycolytic and energy metabolic pathway markers (using novel metabolomic analysis) in microgravity versus 1G. Most importantly, this comprehensive project will lead to the first direct in vitro assessment of canonical fertility-critical functional processes of mammalian sperm in microgravity that are identical to many human in vitro clinical and research laboratory assessments. These experiments are 1) part of the Rapid Turn-around Space Flight Experiments Research Emphasis, and 2) address the National Research Council (NRC) 2011 Decadal Survey to discover and characterize underlying mechanisms of adaptation to the space flight environment, and to determine cellular and organismal mechanisms that regulate and sustain reproduction and development in space. In this regard, the results of these experiments will provide important fundamental reproductive biological data to support NASA's goal to conduct multigenerational studies of mammals in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous studies indicate that reproductive functions in sperm that occur on the ground also occur in space flight. However, the timing of signal transduction changes upon initiation of sperm motility is more rapid, yet, the fertility-related responses of sperm to the presence of egg chemotactic factors are delayed. This suggests that sperm fertility may be affected resulting in subfertility or infertility in males during and post flight. There is insufficient data available from humans to assess this possibility due to low n's and HIPAA concerns.
Task Progress:	November 2022 Update (Ed., 11/29/22) NASA Award #: 80NSSC18K0522 - Final Report PI: V. Gustavo Blanco, M.D. Ph.D. Institution: University of Kansas Medical Center Multigenerational survival beyond Earth is a target research area of the NASA Decadal Survey. Our Micro-11 experiment is the first assessment of live human and bovine sperm functions in space flight using multiple fertility biomarkers that are used clinically on Earth. Multi-generational survival of humans (and animals) require healthy functional sperm and eggs on Earth, in space flight, on potential habitats on the Moon, Mars, or beyond, and after return to Earth. On Earth, sperm 5 function is routinely determined to assess male fertility and prior to applying any assisted reproduction technology (ART), including in vivo insemination (IUI), in vitro fertilization (IVF) or intracytoplasmic sperm injection (ICSI). Sperm capacity to fertilize eggs requires that the cells undergo critical changes (capacitation), which involve the acquisition of a specific pattern of sperm motility, and an exocytotic process in the sperm head called the acrosome reaction (AR). Quantitative assessments of motility and AR to live sperm on Micro-11 (SPX-14) using clinical procedures during space flight provided foundational data on prospects for long-term multigenerational survival in space.
Bibliography Type:	Description: (Last Updated: 04/23/2021)